

RE07 Global Infrastructure Investment and Management

Week 6

Performance and Valuation

Professor Helen X. H. Bao

Department of Land Economy, University of Cambridge

Guest session: Saïd Business School, University of Oxford

What this week does

Every previous week produced numbers: an allowed return, a rating, a fee, a risk allocation. This week asks whether any of them describe what actually happens. It has two halves that do not sit comfortably together. The first is the machinery of project appraisal, which is precise, teachable and used everywhere. The second is the evidence that the forecasts the machinery operates on are systematically wrong, in a known direction, by a known amount.

A	What are we measuring, and for whom	25 min
B	Building the cash flow	35 min
C	Capital structure and debt sizing	30 min
D	Cover ratios and what they hide	45 min
E	Sensitivity, scenario and value at risk	35 min
F	Forecast accuracy and the outside view	40 min
G	Benchmarking, wider value and the limits of a single number	35 min

What you already have

Where this week draws on modules you have taken, or are taking now. Nothing below is assumed: each idea is stated in full before the module is named.

Your programme	Where it was taught	What it gives this week
Real Estate Finance	RE02 Lent lecture 3, valuation methods, and lecture 4, inputs, testing and machine learning. Michaelmas lecture 5, key metrics and property indices. RE01 session 5, indices and price dynamics; session 3, the weighted average cost of capital.	Discounted cash flow, comparables and residual valuation, and the appraisal smoothing problem, all of which transfer to an asset class with longer horizons and almost no comparables.
Planning, Growth and Regeneration	PGR02 Michaelmas week 2, the evaluation of urban policy. PGR05 lecture 4, regional accessibility and transport investment, and block B on evaluating policy tools. PGR03 lectures 5 and 6, agglomeration and spatial equilibrium.	Additionality, deadweight and displacement, and the theory behind the wider economic impacts that appear in a transport business case.
Environmental Policy	EP02, valuing environmental goods and services, and evaluation of environmental policies. EP06, levelised cost of energy and multicriteria decision analysis.	How to value what markets do not price, and how to compare options on more than one objective when a single financial metric will not do it.
Everyone	RM02 topic 6, experiments and quasi-experiments; topic 8, assessing statistical studies.	Reference class forecasting is an outside-view estimator, and internal and external validity is the language for saying why it works.

Block A

What are we measuring, and for whom

Four different questions that all get called performance.

Indicative length 25 minutes

Four questions, four answers

Question	Asked by	Metric	Where it is decided
Can the project service its debt	Lenders	Cover ratios under a downside case	At financial close, and at every test date after
Does the equity earn its required return	Sponsors and investors	Equity internal rate of return, and multiple	On exit, or never
Is this the best use of public money	The state	Benefit-cost ratio and net present value	At appraisal, in principle
Did it deliver what was claimed	Everybody, eventually	Outturn against forecast	Rarely, because evaluation is not funded

The same project, three verdicts

A toll road can service its debt comfortably, deliver a poor return to its equity, and generate substantial economic benefit through time savings that accrue to users who pay nothing towards it. All three statements can be true simultaneously and they are answers to different questions.

This is not a technicality. It explains why a project can be defended by its promoters and attacked by its investors at the same time, and why arguments about infrastructure performance are so often conducted at cross purposes. The discipline is to state which question is being answered before producing a number.

Whose money, and over what period

Two further specifications are needed before any metric means anything. Whose money: the project's, the equity's, or the state's, since a project internal rate of return and an equity internal rate of return differ by the effect of leverage and can point in opposite directions. And over what period: to the end of the concession, to the end of the debt, or to an assumed exit in year ten, since the terminal value assumption frequently carries more of the answer than the operating cash flows do.

A metric without both specifications is not wrong, it is meaningless, and the group project will be marked on whether they are stated.

Block A discussion

Seminar questions

1. A project has a debt service cover ratio of 1.4 and an equity internal rate of return of six per cent. Who is happy, who is not, and what should happen?
2. Under what circumstances would a state proceed with a project that no private investor would fund, and be right to do so?
3. Terminal value can be half the present value of a thirty-year project. What would you require before accepting a terminal value assumption?

Block B

Building the cash flow

Inputs, timing, and the assumptions that do the work.

Indicative length 35 minutes

What goes into a project cash flow

- The timing of the investment, since a construction programme spread over four years discounts very differently from one spread over two.
- Initial investment cost, including contingency, development cost, financing cost during construction and value added tax timing.
- Revenue, from the sales or offtake contract, with its price mechanism, indexation and volume assumptions stated separately.
- Operating expenditure, and separately the lifecycle and major maintenance profile, which is lumpy and frequently understated.
- Working capital movements, which matter more than students expect where receivables from a public counterparty are slow.
- Taxes, including the treatment of depreciation and interest, and any tax losses carried forward from the construction years.
- Macroeconomic variables: inflation, interest rates and, where relevant, exchange rates, each of which should appear once and feed everything.

Real, nominal, and the mistake everybody makes once

A model must be consistently real or consistently nominal. Mixing them, most often by inflating revenue and leaving operating cost flat, produces a margin that improves every year for no reason and a valuation that is simply wrong.

The practical difficulty is that different parts of a project are indexed differently: a tariff may be linked to one index, a maintenance contract to another, and debt service may be fixed in nominal terms. Modelling that correctly requires each line to carry its own indexation, which is why real models are built in nominal terms with explicit indices rather than in real terms with a single deflator.

Cash available for debt service is the number that matters

The metric that drives everything in this week is cash available for debt service: operating revenue less operating cost, tax and maintenance capital expenditure, and before any payment to lenders or shareholders.

It is not profit. Depreciation is not deducted because it is not a cash cost, and interest is not deducted because it is what the number is being tested against. Getting from an accounting profit to cash available for debt service is the single most common arithmetic error in student work and in the group project, and it is worth doing once on the board line by line.

Where optimism enters a model, quietly

- Construction cost, through a contingency set at a level that reflects negotiation rather than the distribution of outcomes.
- The construction programme, since delay costs interest during construction and pushes revenue back a year.
- Ramp-up, the period after opening in which demand builds, which is routinely assumed to be shorter than it is.
- Operating cost escalation, often assumed at general inflation when labour and energy costs move faster.
- Lifecycle and major maintenance, which sits far enough in the future to be discounted into insignificance and is then real when it arrives.
- Terminal value, where an exit multiple is assumed on the basis of today's market conditions applied to a decade hence.

Why a project model needs all three statements

- The cash flow statement is what lenders and shareholders are actually paid from, and it is the basis of every ratio in this week.
- The profit and loss account determines taxable profit, and tax is a cash cost, so the accounting drives the cash.
- The balance sheet is the discipline: if it does not balance, something in the other two is wrong, which is why model auditors check it first.
- Reserve accounts, retentions and distributions can only be represented properly across all three.
- A model built on cash flow alone will misstate tax, and misstating tax misstates the cover ratio, which misstates the debt.

Interest during construction, and why the model is circular

A project borrows to build, and pays interest on what it has borrowed before any revenue exists. That interest is capitalised and becomes part of the amount borrowed, which increases the interest, which increases the amount borrowed.

The circularity is genuine and is resolved iteratively. Its practical significance is that construction delay is more expensive than it appears: an extra six months adds not only cost escalation and deferred revenue but a further period of capitalised interest on the whole drawn amount. This is one reason the week 1 evidence on schedule overruns matters financially as well as reputationally.

Tax, and the three things that matter

- Depreciation timing, since accelerated allowances defer tax and can be worth a great deal in present value on a capital-intensive asset.
- Interest deductibility, which is the tax shield that makes leverage valuable, and which many jurisdictions now cap as a proportion of earnings.
- Losses carried forward from construction and ramp-up, which shelter early operating profit and change the timing of cash tax substantially.
- Value added tax on construction, which is recoverable but with a timing difference large enough to require its own facility.
- Tax is where a model most often diverges from reality, because it is jurisdiction-specific and the modeller is usually not a tax specialist.

Block B discussion

Seminar questions

1. Take the six places optimism enters. Which would you test first if you had time to test only two?
2. A model shows operating margin improving every year. What would you check?
3. Why is cash available for debt service calculated before interest but after tax, when tax depends on interest?

Block C

Capital structure and debt sizing

How much debt a project can carry, and who decides.

Indicative length 30 minutes

Debt is sized, not chosen

In a corporate context management chooses a capital structure. In project finance the debt quantum falls out of the cash flow: lenders specify a minimum cover ratio, apply it to the projected cash available for debt service under an agreed downside case, and the resulting debt service capacity, discounted, is the amount they will lend.

That inverts the usual logic. The sponsor does not decide how much to borrow. It decides how much contracted cash flow to create, and the debt follows. This is why so much sponsor effort goes into the offtake agreement, and why an uncontracted project is financed at a fraction of the leverage.

Two views of the optimal structure

The sponsor's optimum

- Maximise leverage, because equity is the expensive money and debt is cheap
- Every additional unit of debt raises the equity return while contracted cash flow covers service
- Constrained by what lenders will accept and by the risk of a lock-up blocking distributions
- Also constrained by the tax shield, which stops adding value once taxable profit is exhausted

The lender's optimum

- Enough equity to give the sponsor something to lose, which is the alignment argument
- Cover ratio headroom sufficient to absorb the downside case without a covenant breach
- A debt tail: debt maturing well before the concession ends, so there is time to recover if things go wrong
- A structure that survives the sponsor walking away, which is what step-in and share security deliver

The sizing arithmetic, in one paragraph

Take projected cash available for debt service in each period under the agreed downside case. Divide by the required minimum cover ratio to give the maximum debt service the project can support in that period. Build a repayment profile that fits within that envelope, typically sculpted so that repayments follow the shape of the cash flow rather than being level. Discount the resulting debt service at the cost of debt to give the maximum principal.

The two inputs that determine everything are the downside case and the required ratio, and both are negotiated rather than calculated. That is the point at which this technical block becomes a bargaining problem.

Gearing falls over the life of a project, and the discount rate should follow

A project financing begins at seventy or eighty per cent debt and amortises to zero. The weighted average cost of capital therefore rises steadily over the life of the asset as the cheap money is repaid, and a valuation using a single constant rate is an approximation.

For most purposes the approximation is tolerable and the convention is to use it. For a long concession with a heavily front-loaded debt profile it is not, and the correct treatment is either a time-varying discount rate or an adjusted present value approach that values the ungeared project and the tax shield separately. Students who have taken RE01 have the second method already.

The cost of equity for a project company

A project's equity is a residual claim on a highly geared single asset with no diversification, which makes it a great deal riskier than the sponsor's own equity. Estimating its cost is genuinely difficult: there is no traded price, no comparable company with the same gearing and the same single asset, and the risk changes character sharply at completion.

The practical approaches are to observe required returns on comparable transactions, to build up from a risk-free rate with explicit premia for construction, country and illiquidity, or to work backwards from what sponsors have actually accepted. All three are unsatisfying, and stating which one has been used, and its weaknesses, is what an honest appraisal looks like.

Block C discussion

Seminar questions

1. If debt is sized off a downside case, who chooses the downside case, and what stops it from being chosen conveniently?
2. Sculpted repayment fits debt service to the cash flow. What risk does that create that level repayment does not?
3. Why do lenders want a debt tail, and what does the length of the tail tell you about their view of the project?

Block D

Cover ratios and what they hide

The debt service and loan life cover ratios, and the discussion Gatti gives them.

Indicative length 45 minutes

The two ratios

Ratio	Definition	What it tells you	Typical requirement
Debt service cover ratio	Cash available for debt service in a period, divided by debt service due in that period	Whether the project can pay this year	Above one by a margin that depends on the revenue model
Loan life cover ratio	Present value of cash available for debt service to loan maturity, divided by outstanding debt	Whether the project can pay over the life of the loan	A higher multiple, reflecting cumulative capacity
Project life cover ratio	As above, but to the end of the concession rather than the loan	The cushion beyond the debt tail	Higher again, and used to size the tail

What the ratios can tell you, and what they cannot

Gatti puts this well. A cover ratio is a summary of a distribution reduced to a single number under one set of assumptions. It tells you the margin by which the base case covers debt service. It does not tell you the probability of default, because it says nothing about the variance of the cash flow around the base case.

Two projects with identical cover ratios can have very different default probabilities if one has contracted revenue and the other has traffic risk. That is why required ratios differ by sector, and it is why a ratio should always be read together with the sensitivity analysis rather than instead of it.

The certainty equivalent interpretation

Gatti offers a reading that makes sense of the practice. Requiring a cover ratio above one is equivalent to applying a haircut to the expected cash flow, and treating the reduced figure as a certain amount. A ratio of 1.3 is a statement that lenders will lend against roughly three quarters of the expected cash flow, treating that portion as certain.

Read that way, the ratio is not an arbitrary convention but a certainty equivalent, and the difference in required ratios between an availability project and a merchant one is a statement about how much of each cash flow the lenders regard as certain. It also explains why lenders and sponsors argue about ratios rather than about discount rates: in this market, risk adjustment is done in the numerator.

The mechanisms that hang off the ratio

- Lock-up. If the ratio falls below a threshold, distributions to shareholders are blocked and cash accumulates in the project.
- Default. If it falls further, an event of default occurs and lenders acquire rights up to and including acceleration and step-in.
- Cash sweep. If the ratio is strong, surplus cash may be applied to prepay debt rather than released, which deleverages faster.
- Margin ratchet. Pricing may step down as the ratio improves and up as it deteriorates.
- The distance between lock-up and default is the equity's room to manoeuvre, and it is one of the most heavily negotiated numbers in the term sheet.

The ratio inherits the base case

A cover ratio calculated on an optimistic forecast is an optimistic ratio. Nothing in the arithmetic detects bias in the inputs, and the discipline of computing it precisely can create a misleading impression of rigour.

This is the joint between the two halves of the week. Block D gives students a precise instrument. Block F gives them the evidence that the quantity the instrument measures is systematically overstated. Holding both at once, and knowing what to do about it, is the intended learning outcome of the module's technical strand.

Historic, forward-looking, and the difference between them

- A backward-looking ratio measures what happened in the period just ended, and is what a compliance certificate reports.
- A forward-looking ratio measures projected cash against projected service, and is what tells you whether a problem is coming.
- Facilities usually test both, and the forward-looking test is the one that bites first in a deteriorating project.
- The forward-looking test requires an updated projection, which the borrower prepares, which returns the whole discussion to who controls the assumptions.
- That is not a small point. A covenant tested against a forecast produced by the party being tested is weaker than it looks.

What actually happens at a breach

A covenant breach is not a foreclosure. In the ordinary case, a lock-up test failure blocks distributions and the project continues, accumulating cash. A default test failure gives lenders rights they usually choose not to exercise immediately, because accelerating a project loan produces an asset nobody wants to own.

What follows is a negotiation: a waiver in exchange for a fee and tightened terms, an equity cure in which shareholders inject cash to restore the ratio, a restructuring of the repayment profile, or in the limit a change of control. The observed loss on project finance debt is therefore the outcome of a bargaining process, which is why loss given default is so much lower than the security position alone would suggest.

Regulated businesses use different ratios, for a reason

A regulated network is not financed as a project. It is a going concern with an indefinite life and a rolling asset base, financed with corporate debt at the operating company and often further debt above it. The relevant metrics are gearing against the regulatory asset base, interest cover, and the ratio of funds from operations to net debt, and the regulator itself takes a view on all three.

The contrast is instructive. A project is financed against a finite contracted stream and amortises to zero; a regulated utility is financed against a perpetual regulated stream and never repays. Students should be able to say why the same asset type is financed in these two quite different ways, and the answer is the presence or absence of an end date.

Block D discussion

Seminar questions

1. Two projects have a debt service cover ratio of 1.30. One is an availability PPP and one is a toll road. Are they equally safe? What would you want to know?
2. Interpret a required ratio of 1.25 as a certainty equivalent. What proportion of the cash flow are lenders treating as certain, and does that seem right for a contracted renewable project?
3. The lock-up threshold is set at 1.15 and default at 1.05. Who benefits from widening that gap, and who from narrowing it?

Block E

Sensitivity, scenario and value at risk

From one number to a distribution, and what the step buys.

Indicative length 35 minutes

Which variables to test, and why those

- Construction cost and construction period, because they are the largest single source of variance and the evidence says they are biased.
- Revenue volume, and separately revenue price, since they have different drivers and may be correlated with each other.
- Operating cost, including the escalation assumption rather than only the level.
- Inflation, which affects revenue, cost and often debt service differently and therefore does not cancel.
- Interest rates, on any unhedged portion and at refinancing.
- The variables worth testing are those to which the ratio is most sensitive and about which least is known. Testing a variable that is well known and barely matters is a common way of appearing thorough.

Sensitivity and scenario are not the same thing

Sensitivity analysis

- Move one variable at a time and observe the effect
- Identifies which assumptions matter, which is genuinely useful
- Ignores correlation, so the combinations tested may be impossible
- Produces a fan of outcomes with no probability attached

Scenario analysis

- Move a coherent set of variables together to describe a world
- Captures correlation, at the cost of testing only a few worlds
- The lenders' downside case is a scenario, negotiated rather than estimated
- Still produces no probability, unless the scenarios are weighted

Monte Carlo, and what it adds

Gatti and colleagues set out how simulation can produce a value at risk estimate for a project financing: specify distributions for the key variables, specify their correlations, simulate the cash flows many times, and read the resulting distribution of outcomes rather than a single number.

The method became important because the Basel framework offered banks a choice between standardised capital requirements for project finance and internal models, and an internal model requires a distribution. The gain is a probability statement rather than a fan of scenarios. The cost is that the output is only as good as the input distributions, which for a novel asset are themselves assumptions, and a simulation can dress a guess in a confidence interval.

From distribution to loss

- Expected loss is probability of default multiplied by exposure at default multiplied by loss given default, and each term behaves differently in project finance.
- Probability of default varies sharply over the life of the project, being concentrated in construction, which is why a single annualised figure misleads.
- Loss given default is unusually low, because lenders take over a going concern rather than selling an asset, as the recovery evidence in week 3 showed.
- Whether loss given default should be modelled as deterministic or stochastic is a genuine question, and Gatti treats it as such.
- Restructuring rather than enforcement is the normal outcome, which means the observed loss depends on a negotiation as much as on the asset.

Block E discussion

Seminar questions

1. A model shows the debt service cover ratio falling below one only if traffic is thirty per cent below forecast. Given the week 1 evidence, how reassured are you?
2. A simulation gives a one per cent probability of default over the loan life. What would you need to know before believing it?
3. Should the lenders' downside case be a scenario chosen for its plausibility or a percentile of a distribution? What changes if you switch?

Block F

Forecast accuracy and the outside view

The evidence that the numbers are wrong in a known direction, and what to do about it.

Indicative length 40 minutes

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Transport infrastructure projects worth about ninety billion dollars, analysed for the difference between the cost estimate used to decide and the outturn.

Flyvbjerg, Holm and Buhl found cost underestimation in nearly nine out of ten projects, with average overruns of about forty-five per cent for rail, thirty-four per cent for fixed links such as bridges and tunnels, and twenty per cent for roads.

The distribution matters as much as the average. Underestimation is not symmetric error around a correct central estimate; it is a one-sided bias that has not improved over the decades for which records exist.

Their conclusion is that the pattern cannot be explained by error and is best explained by strategic misrepresentation, which is to say that the numbers used to decide are not the numbers the promoters believe.

Demand forecasts

106%

Average overestimate of passenger numbers in rail projects, across 210 projects in fourteen nations.

The revenue side is worse than the cost side. For nine out of ten rail projects, passenger forecasts were overestimated, by an average of a hundred and six per cent. For roads the picture is different: forecasts are inaccurate rather than biased, with half of projects differing from forecast by more than twenty per cent in either direction.

The asymmetry is informative. Rail schemes require political advocacy in a way that road schemes often do not, and advocacy produces one-sided error.

For this week the implication is direct. A cover ratio computed on a rail passenger forecast is computed on a number that has historically been roughly double what materialised.

The challenge to the account

Love, Ahiaga-Dagbui and Irani dispute the strategic misrepresentation explanation. They argue for a probabilistic theory of causation, in which overruns arise from the interaction of many small factors within a complex delivery system, including scope change, design error, ground conditions and coordination failure, rather than from deception at the point of approval.

The distinction matters because the remedies differ. If the cause is systemic complexity, the answer is better delivery practice and more realistic contingency. If it is misrepresentation, better practice will not help, because the promoter already knows the true number. The evidence is not conclusive and the two are not mutually exclusive, which is the honest position to leave students with.

Reference class forecasting, and how to actually do it

The remedy that follows from the bias account is the outside view. Rather than building an estimate from this project's components, identify the class of projects to which it belongs, establish the distribution of outturns against forecasts for that class, and adjust this project's estimate by the appropriate percentile of that distribution.

It is deliberately crude. It ignores everything specific about the project, which is exactly why it works: the specific information is what the inside view already used to produce the biased estimate. United Kingdom appraisal guidance embeds the method as optimism bias uplifts, and the individual coursework applies it to a cover ratio.

The coursework exercise, stated now

- Take the base case model for the case project and compute the debt service and loan life cover ratios as presented.
- Identify the reference class for the project and state the evidence you are using for the distribution of outturns.
- Apply an outside-view adjustment to capital cost and to revenue, stating the percentile you have chosen and why.
- Recompute the ratios and report the effect, including whether the project would have breached lock-up or default in any period.
- State what the sponsor should do differently at financial close in light of your result, in no more than a page.

The planning fallacy, and why the inside view is so persuasive

The psychological finding behind reference class forecasting is that people asked to estimate a task's duration or cost focus on the specifics of the case in front of them and systematically underestimate, even when they know that similar tasks have overrun.

What makes the inside view compelling is precisely its richness: the estimator knows this project, its team, its site and its programme, and all of that detail feels like information. The outside view discards it deliberately, which feels like ignoring what you know. Students should expect to find the method uncomfortable, and should notice that the discomfort is the bias arguing back.

Building a reference class in practice

- Define the class narrowly enough to be comparable and widely enough to have observations: urban light rail rather than transport, offshore wind rather than renewables.
- Collect forecast and outturn pairs, which is the hard part, since forecasts are often revised and the original is not archived.
- Compute the distribution of the ratio of outturn to forecast, not the average error, because the distribution is skewed.
- Choose a percentile appropriate to the decision: a promoter deciding whether to bid and a lender deciding whether to lend want different points.
- State the class, the source and the percentile in the output, so that a reader can disagree with the choice rather than with the result.

What good post-completion evaluation would look like

It would compare outturn cost, schedule, demand and benefit against the figures used to decide, not against a revised baseline. It would be conducted by a body with no stake in the answer. It would be published whatever it found. And it would be fed back into the reference class used for the next appraisal, which is the only mechanism by which the evidence base improves.

Almost nowhere does all four. The most common failure is the second baseline: a project assessed against a budget revised after the difficulties emerged appears to have performed adequately, which is how a system with systematic overruns can report that most projects were delivered to budget.

Block F discussion

Seminar questions

1. Flyvbjerg says misrepresentation, Love says complexity. Design a study that would distinguish them, and say why nobody has run it.
2. If lenders know demand forecasts are biased, why do they lend against them at all? What are they actually relying on?
3. An outside-view adjustment would make many projects fail their cover ratio test. What happens to the market if everybody applies one?

Block G

Benchmarking, wider value and the limits of a single number

Three disciplines, three answers, and the case for not choosing between them.

Indicative length 35 minutes

The benchmarking problem, stated

To say a fund performed well you need something to compare it with. For unlisted infrastructure that comparator does not properly exist. The available indices are built from appraisals rather than transactions, cover short periods dominated by one interest rate cycle, mix revenue models that behave quite differently, and are frequently constructed by parties with an interest in the result.

The serious attempt at a solution is the work of research institutes building valuation models calibrated to observed transactions and applying them consistently across a defined universe. It is a genuine improvement and it does not remove the underlying difficulty, which is that the assets rarely trade.

The wider impacts question

Curse or cure

Flyvbjerg's framing of whether megaprojects deliver benefits beyond their direct users.

Beyond the direct revenue and the direct user benefits sits a set of wider effects: agglomeration, land value uplift, labour market access, regeneration, and the reorganisation of activity in space.

These are real, they are the main justification offered for many large schemes, and they are the least reliably estimated component of any business case. Some of what is claimed as a wider benefit is displacement, activity moving from one place to another with no national gain, which is the distinction the planning cohort met as additionality and deadweight.

The theory behind the estimates is the agglomeration material in PGR03, and the elasticity applied to effective density often determines whether a scheme passes.

Three disciplines, one project

Question	Discipline	Instrument	Blind spot
Can it service its debt	Finance	Cover ratios under a downside case	Says nothing about whether the project was worth building
Did it add activity, net	Planning and evaluation	Additionality, deadweight, displacement	Hard to identify a counterfactual, and rarely funded
What were the non-market effects worth	Environmental economics	Stated and revealed preference valuation	Wide confidence intervals, and contested transfer of values
Are we comparing like with like	All three	A benchmark, or a multicriteria framework	None of the three has a complete answer

When one number will not do

Where objectives conflict and cannot be reduced to a common unit, the honest instrument is multicriteria decision analysis: state the criteria, score the options against each, make the weights explicit, and show how the ranking changes as the weights change.

EP06 teaches this for energy policy and it applies directly to the syllabus phrase about assessing the long-term value and impact of projects. Its virtue is not that it produces a better number but that it makes the value judgements visible instead of burying them in a discount rate or a shadow price. For a project with a financial return, a carbon consequence and a distributional effect, that visibility is worth more than false precision.

Levelised cost is a cover ratio's cousin

The levelised cost of energy takes the lifetime cost of a generating asset, divides by its lifetime output, and expresses the result as a cost per unit. Like a cover ratio it compresses a long and uncertain stream into one comparable number, and like a cover ratio its answer is determined by the discount rate and the assumed load factor rather than by anything visible in the output.

Comparing the two is a useful exercise for a mixed cohort, because the environmental policy students know the pitfalls of one and the finance students the pitfalls of the other, and the pitfalls are the same.

Appraisal smoothing, stated formally

A valuer setting this quarter's appraisal anchors on the last one and moves partway towards current evidence. The reported series is therefore a weighted average of true values across periods, which reduces measured variance and measured correlation with any contemporaneous series, and introduces autocorrelation that true returns should not have.

The standard correction, developed in real estate, is to unsmooth the series by inverting the assumed updating rule, which raises estimated volatility substantially. Applying it to unlisted infrastructure removes much of the apparent risk advantage over listed exposure. Students who met this in RE01 or RE02 should be asked to state it, because it is exactly the same problem in a different asset.

Value for money, revisited with the machinery in place

Week 2 introduced the public sector comparator and noted that the answer is determined by the risk adjustment. With the machinery of this week in place the criticism can be made precisely.

The comparator compares a public option, costed at a base case with a risk adjustment, against a private option priced by a market that has applied its own view of the same risks. If the public base case carries the optimism documented in block F and the private price does not, the comparison is not between two delivery routes but between a biased forecast and a market price. That is a serious methodological problem and it points in a predictable direction.

Distribution, and the question a single metric cannot answer

- A net present value tells you whether the gains exceed the losses. It does not tell you who gains and who loses.
- Infrastructure has strong distributional consequences: a transport link can raise national output and leave the smaller of the two places it connects worse off.
- Tariffs are distributional instruments, and the choice between a user charge and general taxation is a choice about who pays.
- United Kingdom appraisal permits distributional weighting, and it is used sparingly, which is itself a decision.
- Making the distributional consequence explicit does not resolve it. It moves the decision from an analyst to a decision-maker, which is where it belongs.

Block G discussion

Seminar questions

1. A project has a cover ratio of 1.4, a benefit-cost ratio of 1.1, and displaces activity from a neighbouring town. Should it be built, and who decides?
2. Design a benchmark for unlisted infrastructure that meets the five conditions from week 3. What is the binding constraint?
3. Would you rather have a precise financial metric with a biased input, or an imprecise multicriteria assessment with honest weights?

Block H

Reserve material

Slides held back from the main sequence.

Indicative length As required

The case that runs the numbers end to end

Gatti case study 2, the Italy water system, is the only case in the set text with a complete business plan: timing, inflation, depreciation, interest, taxation, working capital, a debt service reserve, capital expenditure, the sources of financing including a public grant, the operating period across the existing system, new supply and treatment works and two hydroelectric plants, and the economic and financial ratios that result.

It pairs with the appendix, which walks through the structure of a real model section by section: assumptions, capital expenditure, sensitivities, lifecycle costs, value added tax, depreciation, the public grant, energy and water revenues, operating cost, the three statements, tax, working capital, the guarantee facility, debt, equity and the resulting internal rate of return.

An alternative running order

Where the cohort is quantitative, compress blocks A and B, run C and D with a model open, and give the recovered time to block E, taking the value at risk material in full.

Where the cohort is drawn mainly from Planning and Environmental Policy, keep A and B intact, treat C as background, teach D to the certainty equivalent slide and no further, and give the time to F and G, which are the parts of the week their own disciplines speak to directly.

Sources drawn on for this deck

- Gatti, S. (2024), Project Finance in Theory and Practice, chapter 5, chapter 8 sections 5 to 9, case study 2 and the appendix.
- Delmon, J., Private Sector Investment in Infrastructure, chapters 2 and 15.
- Flyvbjerg, B. (ed.) (2017), The Oxford Handbook of Megaproject Management, chapters 3, 9 and 17.
- Weber, B., Staub-Bisang, M. and Alfen, H. W. (2016), Infrastructure as an Asset Class, chapter 5.
- EDHECinfra material on benchmarking unlisted infrastructure, in the module resources folder.
- Journal articles as cited on each slide.

Where Week 6 leaves us

Performance is four different questions, and most arguments about infrastructure are conducted between people answering different ones.

A cover ratio is a certainty equivalent: a statement about how much of the expected cash flow lenders treat as certain.

The forecasts beneath every ratio are biased in a known direction by a known amount, and the arithmetic cannot detect it.

The outside view is crude, and its crudeness is why it works, because the specific information is what produced the bias.